High-Level Design (HLD)

1. System Overview

Problem Definition

Customer support teams receive repetitive queries about products, troubleshooting, warranty, and policies. Traditional chatbots often hallucinate or provide incorrect answers because they lack access to official documentation. There is also a need to intelligently escalate sensitive or complex queries to human agents.

Scope of the System

The system will:

- Ingest customer support PDF documents
- Convert documents into searchable knowledge using RAG
- Answer user queries with grounded responses
- Use LangGraph for intelligent workflow orchestration
- Support conditional routing and Human-in-the-Loop (HITL) escalation
- Provide a CLI-based interface *(Streamlit optional)*

Out of Scope

- Multi-document support
- Voice interface
- User authentication
- Production deployment

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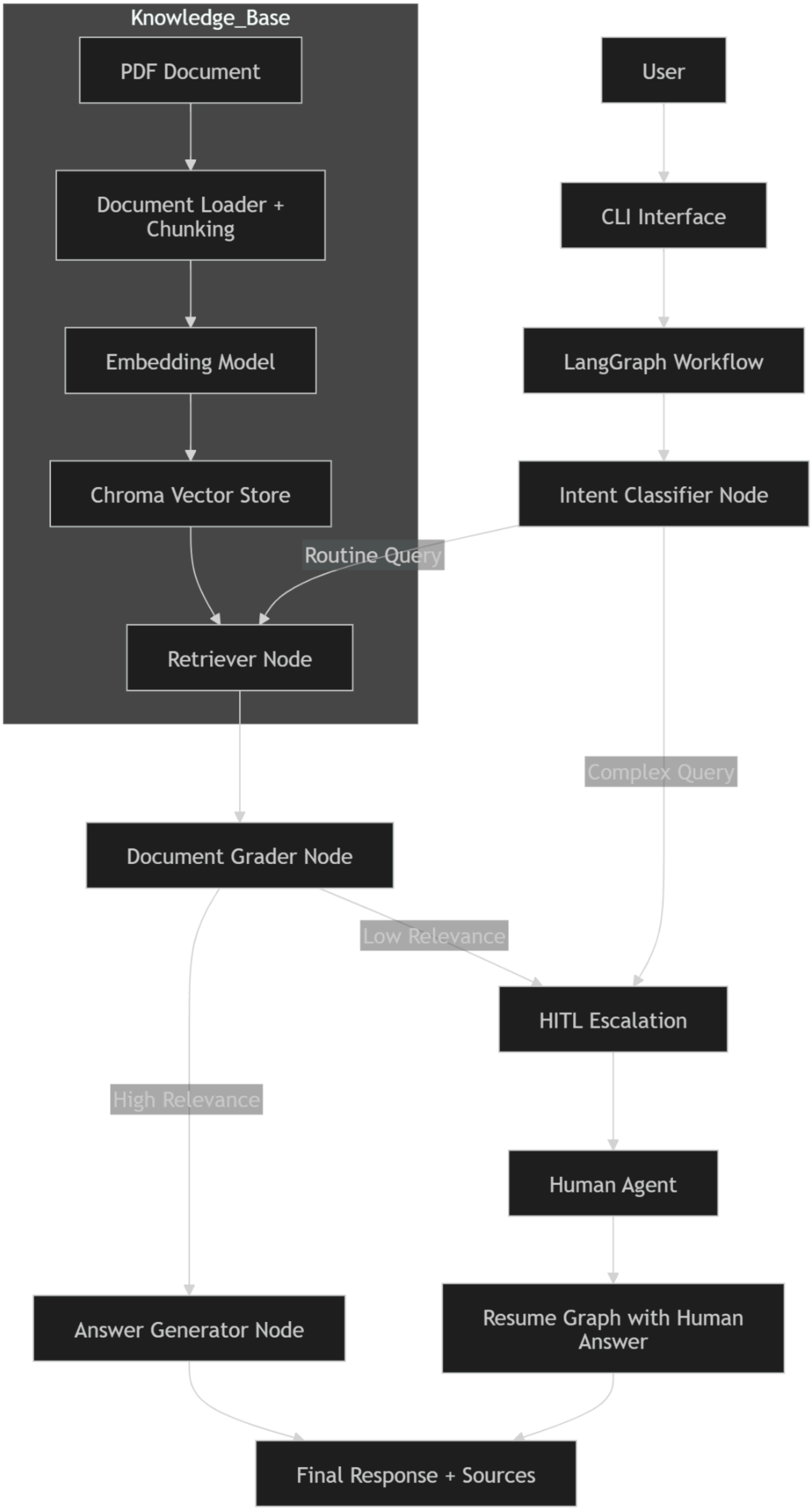
2. Architecture Diagram



Knowledge_Base
PDF Document
Document Loader + Chunking
Embedding Model
Chroma Vector Store
Retriever Node
User
CLI Interface
LangGraph Workflow
Intent Classifier Node
Routine Query
Complex Query
Document Grader Node
Low Relevance
High Relevance
HITL Escalation
Human Agent
Answer Generator Node
Resume Graph with Human Answer
Final Response + Sources

3. Component Description

Component	Description	Technology Used
Document Loader	Loads and extracts text from PDF	PyPDFLoader
Chunking Strategy	Splits documents into chunks with overlap	RecursiveCharacterTextSplitter
Embedding Model	Converts chunks into embeddings	all-MiniLM-L6-v2
Vector Store	Stores and indexes embeddings	ChromaDB
Retriever	Retrieves relevant chunks	Chroma `as_retriever()`
LLM	Classification, grading, generation	zephyr-7b-beta
Graph Workflow Engine	Orchestrates stateful flow	LangGraph (StateGraph)
Routing Layer	Determines next action	Conditional Edges
HITL Module	Human escalation and resume	LangGraph Interrupt + Checkpointer

4. Data Flow

Ingestion Phase (One-Time)

PDF
→ Document Loader
→ Text Chunking
→ Embeddings
→ ChromaDB

Query Lifecycle

User Query
→ Intent Classification
→ Retrieval

→ Relevance Grading

If relevant:
 → Generate Answer
Else:
 → Human Escalation

→ Final Response with Sources

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5. Technology Choices

ChromaDB

- Lightweight and persistent
- Easy local setup
- No separate server required

LangGraph

- Graph-based control flow
- Stateful execution
- Native HITL support

LLM

- zephyr-7b-beta
- Chosen for speed and cost-effectiveness

Additional Tools

- LangChain
- sentence-transformers

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6. Scalability Considerations

Large Documents

Future improvements:

- Hierarchical chunking
- Parent-document retriever

High Query Volume

- Async processing
- Response caching

Latency Optimization

- Smaller embedding models
- Chunk size tuning